CODING & ROBOTICS – GRADE 4
STUDY NOTES & CODE

TERM 1 CONTENT

LESSON 7
Control in Scratch
- Repeat loops in our daily lives.
- Use a wait block in Scratch.

CODE – REPEAT LOOP

```
when [flag] clicked
repeat (10)
    move (10) steps
    wait (1) seconds
```

KEY: repeat – do something again and again.

LESSON 9
Loops and conditional statements
- Forever loops in real-life situations.
- If – then – else statements in real-life situations.

CODE – FOREVER LOOP

```
when [flag] clicked
forever
    move (10) steps
    if on edge, bounce
```

KEY: forever – keep doing something without stopping.

CODE – IF THEN ELSE

```
if <touching (edge) ?> then
    turn ↻ (15) degrees
else
    move (10) steps
```

KEY: if – then – else – make decisions.

LESSON 10
Our digital world
- What is technology? (Technology, non-technology and IT).
- Input, Processing and Output (IPO).

IPO CYCLE

INPUT Data goes in. → **PROCESSING** Data is processed. → **OUTPUT** Information comes out.

EXAMPLES
- Keyboard (input) → Computer (processing) → Screen (output)
- Camera (input) → Phone (processing) → Photo (output)

KEY VOCABULARY

Loop:	A block of code that repeats.
Repeat Loop:	Repeats a block of code a fixed number of times.
Forever Loop:	Repeats forever (until stopped).
Wait Block:	Pauses the code for some time.
If – Then – Else:	A conditional statement used to make decisions.
Technology:	Things that solve problems and make life easier.
IT:	Information Technology.
Input:	Data entered into a device.
Processing:	Data is worked on.
Output:	Information produced.
Variable:	A storage box for data.
Operator:	A symbol used to perform operations.
Pen Function:	Used to draw with the sprite.
Robot:	A machine that can do tasks automatically.

TERM 2 CONTENT

LESSON 11
Sensing in Scratch
- Colour-sensing blocks.

CODE – COLOUR SENSING

```
if <color [ ] is touching ( ? )> then
    say (Blue colour!) for (2) seconds
```

KEY: Detect colours and respond to them.

LESSON 12
Variables and Operators
- What are variables?
- What are operators?

CODE – VARIABLE

```
set [score] to (0)
change [score] by (1)
say (score)
```

CODE – OPERATORS

```
(5) + (3)
(10) - (4)
(3) * (2)
(10) / (2)
pick random (1) to (10)
(50) > (20)
```

LESSON 14 AND 15
Use the pen function.
- What are pens, why do we need them, etc.
- Pen code blocks.
- Use the pen function in Scratch.
- Draw a hexagon in Scratch.

PEN CODE BLOCKS

```
pen down
pen up
set pen color to ( )
set pen size to (2)
erase all
stamp
```

CODE – DRAW HEXAGON

```
when [flag] clicked
pen down
repeat (6)
    move (100) steps
    turn ↻ (60) degrees
pen up
```

LESSON 16
Robots and their impact on the world.
- What is a robot, what job does it do, etc.

EXAMPLES OF ROBOTS
- Industrial Robot – used in factories.
- Medical Robot – used in surgeries.
- Domestic Robot – used for cleaning, etc.
- Space Robot – used for exploration.

SCRATCH CODE QUICK REFERENCE

Events	when [flag] clicked
Control	repeat (10); forever
Sensing	touching (edge) ?
Operators	(5) + (3); pick random (1) to (10)
Variables	set [score] to (0)
Pen	pen down

REAL-LIFE CONNECTIONS
- Loops – traffic lights, washing machine cycle.
- If – Then – Else – crossing the road, weather decisions.
- IPO – recipe (ingredients → cooking → food).
- Variables – keeping score in a game.
- Pen Function – drawing shapes and patterns.
- Robots – help humans in many jobs.

EXAM TIPS
- Read the question carefully.
- Use correct Scratch blocks.
- Test your code before submitting.
- Use meaningful variable names.
- Check your project and save your work.

PRACTICE IDEAS
- Make a sprite move in a square using a loop.
- Use if – then – else to change costume.
- Make a colour-sensing game.
- Keep score using a variable.
- Draw different shapes using the pen.
- Research a robot and write its uses.

IMPORTANT REMEMBER
- Practice regularly.
- Understand each concept.
- Use code blocks correctly.
- Think logically and be creative!

KEEP PRACTICING, KEEP CODING!